

A Speedup vs. local PyTorch (GM over cases)

B Change in relative speedup

Point-wise

Reduction/Norm.

MatMul/Attention

Stencil/Conv.

Data Layout

	B200			GH200			MI300X
	Triton	cuTile	TileLang	Triton	cuTile	TileLang	
dequantize_rowwise	5.95	5.16	5.80	5.19	4.61	5.02	6.41
dropout	1.56	1.55	1.57	1.51	1.48	1.52	1.12
fused_activation	1.80	1.78	1.78	1.85	1.83	1.83	1.24
leaky_relu	1.03	1.03	1.02	1.01	1.00	1.01	0.77
mul2	1.09	1.04	1.12	1.04	1.00	1.04	0.82
quantize_global	1.01	1.01	0.98	1.03	1.01	1.02	0.65
relu	1.12	1.06	1.15	1.03	1.01	1.04	0.91
rope	10.21	8.94	10.25	6.70	6.44	6.71	3.93
sigmoid	1.24	0.95	1.21	1.04	0.95	1.03	0.89
swiglu	1.81	1.47	1.69	1.65	1.48	1.64	1.22
vector_add	1.03	1.02	1.06	1.02	0.99	1.02	0.74
weight_dequant	12.63	7.30	13.80	9.24	7.66	9.34	7.03
argmax	1.99	2.06	2.37	1.65	1.58	1.73	0.96
batch_normalization	2.86	2.56	3.27	2.12	1.86	2.07	3.89
cross_entropy	2.49	1.76	2.34	1.85	1.44	1.95	1.94
histogramming	1.21	0.75	4.82	1.25	0.76	5.36	0.61
kl_divergence	2.84	2.97	3.17	3.19	3.14	3.18	2.65
l2_norm	6.00	5.24	6.35	4.05	4.12	4.39	3.08
layernorm	2.00	1.65	2.31	1.40	1.33	1.52	1.17
mean_reduction	1.83	1.61	1.84	1.32	1.30	1.32	0.81
moe_topk_gating	9.56	4.88	9.25	8.73	4.48	9.04	9.65
rmsnorm	1.28	1.19	1.42	1.17	1.18	1.30	1.12
softmax	2.40	1.58	2.14	1.67	1.33	1.61	1.00
batched_matmul	0.81	0.71	0.61	1.02	0.88	0.87	0.29
block_sparse_attention	1.95	0.72	0.71	1.91	0.88	1.11	0.42
flash_attention	0.68	0.72	0.40	0.71	0.83	0.53	0.31
flash_decode	1.98	0.59	1.99	1.89	0.61	1.92	1.28
linear_self_attention	0.56	0.28	0.17	0.54	0.44	0.19	0.27
matmul_fp32_fp16_fp8	0.87	1.10	0.52	0.98	0.61	0.41	0.09
matmul_int8	10.34	10.16	10.88	13.50	6.80	6.86	0.87
streamk_matmul	0.46	0.23	0.30	0.53	0.37	0.46	0.08
1d_conv	0.54	0.31	0.10	0.57	0.46	0.10	1.92
2d_conv	0.24	0.15	0.06	0.25	0.17	0.05	0.53
2d_max_pooling	2.64	2.03	3.36	2.58	2.47	2.28	1.63
3d_conv	0.12	0.05	0.03	0.18	0.09	0.04	1.09
gaussian_blur	13.86	11.41	10.05	13.80	11.11	14.79	5.18
jacobi_stencil_2d	5.78	6.67	6.90	5.62	6.12	6.26	3.22
bitonic_sort	36.43	36.59	36.53	33.40	31.16	33.94	20.09
destindex	3.49	1.67	3.28	2.53	1.54	2.37	1.69
interleave	3.63	3.56	3.49	2.87	2.82	2.74	1.99
matrix_copy	1.08	1.09	1.08	1.04	1.01	1.04	0.86
matrix_transpose	5.70	5.68	4.60	5.16	5.11	4.55	4.11
radix_sort	0.69	0.37	0.39	0.65	0.35	0.37	0.62
reverse_array	2.33	2.31	1.99	1.89	1.80	1.67	1.30
top_k_selection	3.66	3.24	3.45	2.34	3.06	2.42	3.54

